

Pilot underway — sample log request

To get the pilot underway, could you first send us a sample log from one simple, low-speed movement using the manufacturer's existing tools? The operator can hold the robot in a neutral pose for 5–10 seconds, move one joint slowly through a safe, moderate range, hold for 2–3 seconds, return to the starting position, and remain stationary for another 5 seconds. No additional Synexis software is required.

Log contents (ideally)

- timestamps;
- joint name/ID;
- commanded and measured position and velocity;
- torque or motor current if available;
- controller state;
- any warnings, faults or safety events.

Robot & command metadata (please also include)

- robot model;
- controller/firmware and SDK versions;
- logging frequency;
- units;
- joint limits;
- configured speed and acceleration limits;
- payload;
- how the command was issued.

The manufacturer's original log plus a CSV or JSON export would be ideal.

Delivered sample — 2026-09-16 18:12 (America/Los_Angeles)

What was executed (robot moved: forward, then back)

- robot stood up (posture command) — then hold neutral 8.4 s
- move slowly forward — +0.10 m/s for 2.4 s
- hold — 2.1 s
- return (reverse) — -0.10 m/s for 2.5 s
- hold — 4.5 s, then stop (brake frames)

Files (this run)

- pilot_slow_move3_sample.csv — one row per event: ISO + elapsed timestamps; command rows (vx/vy/wz + result); frame rows (code, size, decoded fields)
- pilot_slow_move3_sample.json — same data + full metadata block
- pilot_slow_move3-20260916-181238.hex — raw manufacturer frames (hex) as received

Log summary

- duration 20.0 s · 5 timestamped commands · 115 streamed frames
- streamed frame codes: 0x0F02 (49), 0x0F03 (31), 0x0F01 (30), 0x0906 (5) — low-level telemetry frames of the Lite3 motion host
- commands logged at send time with exact values; inbound frames appended with receive timestamps

Metadata (as collected)

- robot model: DeepRobotics Jueying Lite3
- controller: Lite3 motion host — high-level UDP interface (commands :43893, telemetry :43897)
- logging SDK: lite3sdk (zero-dependency; same protocol family as the manufacturer's high-level SDK)
- logging frequency: as streamed by the robot, receive-timestamped; every command logged per send
- units: SI (m, m/s, rad, rad/s); raw frames also supplied in hex
- limits applied: $v_x \leq 0.5$ m/s, $v_y \leq 0.4$ m/s, $w_z \leq 0.8$ rad/s (test performed at 0.10 m/s)
- how the command was issued: direct UDP velocity frames (0x140 forward/back, 0x145 lateral, 0x141 yaw) with the manufacturer's mode/heartbeat frames, sent from a control laptop on the robot's Wi-Fi

Notes

- An earlier attempt with identical commands produced no motion because the robot was sitting/lying — velocity frames are only honoured while the robot is standing. Cause identified, and the successful run below includes the stand step.
- UDP provides no acknowledgement; motion was confirmed by direct observation during the run.

Still to be supplied (not available on the high-level interface)

- per-joint commanded vs measured position/velocity and torque/motor current at joint level
- controller state and internal warnings/faults
- firmware/Deeprcs and low-level SDK versions, joint limits, acceleration limits, payload

These require the manufacturer's low-level tooling: *Lite3_MotionSDK* (1 kHz RobotData: joint position/velocity/torque, IMU, contact forces) or the *Lite3_ROS* bridge (rosviz of /joint_states, /leg_odom2, /imu/data), run with the robot's ~/jy_exe/conf/network.toml pointing at the logging host. Once that access is available the same manoeuvre will be re-run and the joint-level CSV/JSON added here.